

Bryan GALARZA

Antwerp – Belgium

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🔍 Google Scholar • ResearchGate • GitHub • 🌐 Personal page

Career Profile

Operations Research specialist with a strong background in developing optimization algorithms for both academic and real-world decision-making problems. My work bridges theory and practice, with a focus on metaheuristics, exact methods, and data-driven approaches to support efficient, practical solutions across domains such as transportation, logistics, and healthcare.

⚙️ **Core skills:** Combinatorial Optimization | Metaheuristics | Transportation | Algorithms | Programming
🇧🇪 **Nationality:** Belgian citizen (EU)

Education

University of Antwerp, ANT/OR

Doctor of Philosophy (Ph.D.)

Operations Research

Supervisors: Kenneth Sörensen and Pieter Vansteenwegen

Thesis: Towards the Goldilocks Zone of demand-responsive transportation services [🔗](#)

Antwerp, Belgium

Jun 2019 – May 2023

Ghent University, Industrial Systems Engineering

Master of Science in Engineering Sciences, Cum Laude

Industrial Engineering and Operations Research (burgerlijk ingenieur)

Thesis: Multimodal coordination schemes for Intelligent Traffic Systems [🔗](#)

Ghent, Belgium

Sep 2016 – Jul 2018

Ghent University, Engineering and Architecture

Bachelor of Science in Engineering Sciences

Chemical Engineering and Material Science (burgerlijk ingenieur)

Ghent, Belgium

Sep 2012 – Jul 2016

Sint-Lievenscollege

High-school, general secondary education (ASO)

Science and Mathematics

Antwerp, Belgium

Sep 2005 – Jul 2011

Experience

Department of Engineering Management (ENM) - University of Antwerp

Postdoctoral Researcher

Antwerp, Belgium

Oct 2025 – Present

I work on the STRAUSS project, which focuses on urban logistics within the field of Operations Research, and on healthcare transportation research. My tasks include:

- Developing algorithms and frameworks for solving urban logistics challenges.
- Supervising and supporting PhD students in their research.
- Collaborating with academic and industry partners to ensure practical impact.
- Working on research proposals and publications to advance the field of logistics.

VLAIO - Triptomatic

Lead Operations Research Expert

(remote) Duffel, Belgium

Oct 2023 – Oct 2025

Triptomatic is a software company offering digital solutions in healthcare. I am fully responsible for the Operations Research part of a VLAIO-funded development project. My tasks include:

- Conducting research on on-demand transportation problems in healthcare and translating them into OR models.
- Designing and implementing real-time optimization algorithms for vehicle dispatching and routing.
- Integrating Operations Research algorithms into a decision-support framework with GIS-based tools.
- Performing data analysis to evaluate algorithmic performance and identify systemic improvement opportunities.
- Bridging academic research with industrial application, ensuring scientific rigor while delivering practical solutions.

Department of Engineering Management (ENM) - University of Antwerp

Postdoctoral Volunteer Researcher

Antwerp, Belgium

May 2023 – Oct 2025

As a postdoctoral researcher, I continued to write and publish academic papers in the field of Operations Research. I also presented my research in international conferences. I remained affiliated to the university to pursue further collaborations and prepare grant proposals.

Doctoral Researcher

Jun 2019 – May 2023

My PhD research focused on designing and optimizing semi-flexible, on-demand transportation systems.

Key achievements and tasks:

- Researched and designed novel semi-flexible bus services.
- Developed novel algorithms for static and online real-time optimization of on-demand feeder services.
- Published five academic papers in peer-reviewed international journals.
- Presented at multiple international conferences and gave invited seminars.
- Participated in industry collaboration projects, acting as a consultant for companies.

Atlas Copco

Wilrijk, Belgium

Improvement consultant

Sep 2018 – Apr 2019

Project-based consulting work focused on optimizing packaging policies. My tasks included:

- Documented existing packaging policy for piping components.
- Proposed improvements using linear programming and metaheuristics.

Languages

English: Proficient

Spanish: Proficient

Dutch: Proficient

French: Elementary

Computer skills

Programming:

- *Proficient:* C++, Python, Java, R, MATLAB
- *Basic:* SQL, MongoCxx

Software:

- *Proficient:* L^AT_EX, CPLEX, Gurobi, Hexaly, CI/CD tools (Bitbucket, Github)
- *Intermediary:* Docker, Jira, FlexSim, VISSIM, Maple, AMPL

Microsoft Office / Google workspace: Excel / Sheets, PowerPoint / Slides, Word / Docs, Teams / Meet, Outlook / Gmail, Drive, Calendar (Proficient)

Accomplishments

Sint-Lievenscollege

Antwerp, Belgium

Dr. Splichal Award

2011

Award for the best high-school thesis.

Sint-Lievenscollege

Antwerp, Belgium

Zuster Roes Award

2011

Award for significant improvement in various aspects of high-school education.

University of Antwerp

Antwerp, Belgium

Best Paper Award

2020

Finalist (2nd place) for the *Best Paper Award* in the Doctoral Day of the Faculty of Business and Applied Economics

EURO

Monterrey, Mexico

ELAVIO scholarship

2022

Winner of the EURO scholarship for attending the ELAVIO summer school.

Publications (6)

A large neighborhood search algorithm to optimize a demand-responsive feeder service

Transportation Research Part C: Emerging Technologies, 127 (2021)

DOI: <https://doi.org/10.1016/j.trc.2021.103102>

A survey on demand-responsive public bus systems

Transportation Research Part C: Emerging Technologies, 137 (2022)

DOI: <https://doi.org/10.1016/j.trc.2022.103573>

A column generation algorithm for the demand-responsive feeder service

Networks, 80(3) (2022)

DOI: <https://doi.org/10.1002/net.22095>

The real-time dynamic online feeder service with a maximum headway at mandatory stops

Transportmetrica A: Transport Science, (2023)

DOI: <https://doi.org/10.1080/23249935.2023.2227738>

A demand-responsive feeder service with a maximum headway at mandatory stops

Networks, 83(1) (2023)

DOI: <https://doi.org/10.1002/net.22185>

Towards the Goldilocks Zone of demand-responsive bus services

4OR, 22, PhD Thesis Abstract (2024)

DOI: <https://doi.org/10.1007/s10288-023-00546-4>

Conference presentations (9)

ORBEL 34

Lille, France

34th Annual Conference of the Belgian Operations Research Society

2020

A demand responsive feeder-system service with mandatory and clustered, optional bus stops

ORBEL 35

Virtual

Corona Sessions: Public Transportation

2021

A demand-responsive feeder service with mandatory and optional bus stop

EURO 2021

Athens, Greece

31st European Conference on Operations Research

2021

A large neighbourhood search algorithm to optimize a demand-responsive feeder service

NORS 2021

Bergen, Norway

The Norwegian Operations Research Society

2021

A demand-responsive feeder service with mandatory stops and frequency constraints

ELAVIO 2022

Monterrey, Mexico

Latin Ibero-American Summer School On Operations Research

2022

A demand-responsive feeder service with a maximum headway at mandatory stops

MIC 2022

Ortigia-Syracusa, Italy

14th Metaheuristics International Conference

2022

A demand-responsive feeder service with a maximum headway at mandatory stops

ORBEL 36

Ghent, Belgium

36th Annual Conference of the Belgian Operations Research Society

2022

Towards better service quality with the dynamic feeder service with a maximum headway at mandatory stops

CLAIO 2022

Buenos Aires, Argentina

XXI Latin Ibero-American Conference On Operations Research

2022

Towards better service quality with the dynamic feeder service with a maximum headway at mandatory stops

IFORS 2023

Santiago, Chile

The 23rd Conference of the International Federation of Operational Research Societies

2023

The real-time dynamic online feeder service with a maximum headway at mandatory stops

Hobbies

: Planning and exploring diverse cultures through travel.

: Landscape and urban photography using a mirrorless camera.

: Long-distance hiking, often in mountainous regions.

: Sketching fictional characters and concept art.